	Sanjivani Rural Educational Society's **SANJIVANI COLLEGE OF ENGINEERING** **(An Autonomous Institution)** Kopargaon – 423 603, Maharashtra.
Academic Year: 2025-2026	**T.Y B.Tech Computer Engineering**
	Sample Question Bank –Applied Machine Learning for Industrial Solutions

AMLIS Unit-1 Questions:

1. For each of Supervised, Unsupervised, and Reinforcement Learning, provide a practical example and explain the differences in data requirements and learning approach.
2. Given a probability distribution of events, compute the conditional probability using Bayes' theorem and relate it to a spam email classification example.
3. Given a dataset where the price attribute has minimum and maximum values of 3,000 and 5,000 respectively, and is normalized to the range [0.5–2.0] using min-max normalization, write a Python program to transform the value 7,000 using this normalization technique. Determine through your implementation whether the resulting normalized value falls within or outside the specified range, and discuss any out-of-bound error encountered
4. suppose that the data for analysis includes the attribute age . The age values for the data tuples are (increasing order) 13,15,16,16,19,20,20,21,22,
   22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70.
   i. Use min-max normalization to transformation the value 35 for age with range [0.0,1.0]
   ii. Use z-score normalization to transformation the value 35 for age, where the standard deviation of age is 12.94 years.
   iii.Use normalization by decimal scaling to transform the value 35 for age.

5. Using the given dataset of hours spent driving (x): (10,9,2,15,10,16,11,16) and risk score (y): (95,80,10,50,45,98,38,93), apply Simple Linear Regression to find the equation of the best-fit line that predicts the risk score from driving hours value 9.
6. Implement a linear regression model to predict house prices from synthetic data: size, location, rooms, amenities, price; interpret the slope and intercept values.
7. Given a binary dataset of people's gender (Male / Female) as male or female from their height, then the first class could be male, implement logistic regression and compute the predicted probability for a given input.
8. Explain the working principle of the Support Vector Machine (SVM) algorithm, describing how it determines the optimal separating hyperplane. Include the concepts of margin, support vectors, and the role of kernel functions with a suitable example.

9. A COVID-19 test is conducted on 100 people:

- Actual cases: 30 positive, 70 negative
- Test results: TP = 25, FN = 5, TN = 60, FP = 10 ,Calculate the Accuracy, Precision, Recall, and F1-score for this test.

10. A healthcare company is designing an AI system: Predict patient recovery time, Group patients with similar symptoms, Train a surgical robot. Identify the most suitable ML type for each task and justify.

11. Using the scikit-learn framework, compute the vector addition and dot product for the given vectors a = [4, 3] and b = [2, 5], and interpret the results.

12. Using the scikit-learn framework, for matrices: A = [[1, 2], [3, 4]], B = [[5, 6], [7, 8]] — (a) Perform A × B. (b) transpose of matrix and display the results.

13. Using the scikit-learn framework, implement a Python program to compute and display the gradient vector of the function f(x,y)= f(x, y) = x^2 + 2xy + 3y^2 at a given point (x,y)(*x,y*). Demonstrate the result using at least one non-trivial input point.

14. Using a real or sample dataset containing noisy data, apply binning and clustering techniques to preprocess and improve data quality. Implement these methods using an appropriate data science framework, and display how each technique helps to reduce the impact of noise in your results.

15. Given a dataset containing a 'Salary' attribute with minimum value ₹20,000 and maximum value ₹80,000, and knowing that the mean is ₹45,000 and the standard deviation is ₹10,000, calculate the z-score normalized value for any given salary. Using your program, normalize a sample value (e.g., ₹60,000). Based on your results, explain practical scenarios where z-score normalization is preferred over other techniques.

16. Demonstrate different types of Exploratory Data Analysis (EDA) techniques: such as univariate analysis, bivariate analysis, and multivariate analysis.

17. Explain the theory behind logistic regression as you implement the model, interpret the coefficients, and evaluate the model's performance on the dataset.

18. Given Dataset as Hours Studied: 2, 4, 6, 8; Exam Passed: 0, 0, 1, 1 — Use linear regression to estimate coefficients & Predict for 5 hours studied.

19. A logistic regression model gives P(y=1|x) = 0.72. (a) Classify with threshold = 0.7. (b) Explain recall effect if threshold lowered to 0.5.

20. A quality control test is conducted on 100 manufactured products: Actual defective products: 20 defective, 80 non-defective

Test results: True Positives (TP) = 15, False Negatives (FN) = 5, True Negatives (TN) = 70, False Positives (FP) = 10.

Calculate the Accuracy, Precision, Recall, and F1-score for this quality control test.

21. A new diagnostic tool for detecting a rare genetic disorder was tested on a group of 200 individuals**.**Out of these**,** 50 actually had the disorder (positive class), and 150 did not (negative clas The tool produced the following results:True Positives (TP): 40, False Negatives (FN): 10 ,True Negatives (TN): 135 , False Positives (FP): 15. Apply your understanding of evaluation metrics to compute the following for this diagnostic tool: Accuracy , Precision , Recall (Sensitivity) , F1-score .

AMLIS Unit-2 Questions:

1. Apply PCA to the Iris dataset to reduce features to 2 principal components and visualize the results.
2. Explain the Exploratory Data Analysis (EDA) and its Techniques with examples.
3. Dataset collected in a cosmetics shop showing details of customers and whether or not they responded to a special offer to buy a new lip-stick is shown in table below. Use this dataset to build a decision tree, with Buys as the target variable, to help in buying lip-sticks in the future. Find the root node of decision tree. According to the decision tree you have made from previous training data set, what is the decision for the test data: [Age < 21, Income = Low, Gender = Female, Marital Status = Married]?

id	age	income	gender	marital status	buys
1	< 21	High	Male	Single	No
2	< 21	High	Male	Married	No
3	21-35	High	Male	Single	Yes
4	>35	Medium	Male	Single	Yes
5	>35	Low	Female	Single	Yes
6	>35	Low	Female	Married	No
7	21-35	Low	Female	Married	Yes
8	< 21	Medium	Male	Single	No
9	< 21	Low	Female	Married	Yes
10	>35	Medium	Female	Single	Yes
11	< 21	Medium	Female	Married	Yes
12	21-35	Medium	Male	Married	Yes
13	21-35	High	Female	Single	Yes
14	>35	Medium	Male	Married	No

4. Apply PCA on the MNIST dataset to reduce its dimensionality, visualize the data in 2D, and analyze the impact on model training time and accuracy.
5. Implement Gradient Boosting on a real-world dataset and illustrate how boosting iteratively improves weak learners to build a strong predictive model
6. Using the given ages of website visitors [16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66], apply the K-Means clustering algorithm to group them into suitable clusters and interpret the results.
7. Apply Decision Trees on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.
8. Implement K-Means, Hierarchical Clustering, and DBSCAN on a suitable dataset, compare their clustering results, and interpret the differences in terms of cluster formation and outlier detection.
9. Use histogram and scatterplot visualizations on a dataset to detect patterns, trends, and anomalies, and explain the insights obtained.
10. Apply correlation analysis on a dataset to identify relationships between variables, and interpret the strength and direction of these relationships.
11. Conduct hypothesis testing on a relevant dataset to validate an assumption, and interpret the statistical significance of the results.
12. Apply Single Linkage, Complete Linkage, and Average Linkage hierarchical clustering methods on a sample dataset. Generate dendrograms for each method and compare the clustering structures.
13. Apply Random Forest on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.
14. Apply Gradient Boosting Machines on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.
15. Apply different distance metrics (Euclidean, Manhattan, Minkowski) in a clustering algorithm and analyze how each metric influences the resulting cluster formation.
16. Apply K-Means, Hierarchical Clustering, and DBSCAN on suitable industrial datasets, and analyze their clustering results in the context of real-world applications
17. Implement a Random Forest classifier on a relevant dataset to perform classification, and interpret how the model makes predictions
18. Compare the performance of Gradient Boosting and Random Forest on the same dataset, and analyze differences in accuracy, feature importance, and interpretability
19. Apply K-Means clustering to group customers of a grocery store based on purchase frequency and basket size, and visualize the clusters.
20. Apply the DBSCAN clustering algorithm on a suitable dataset to identify clusters and outliers, and interpret the results.
21. Implement K-Means, Hierarchical Clustering, and DBSCAN on a suitable dataset, compare their clustering results, and interpret the differences in terms of cluster formation and outlier detection.

AMLIS Unit-3 Questions:

1. Implement a simple image classification model using TensorFlow, train it on the MNIST dataset, and evaluate its accuracy.
2. Implement a machine learning model using PyTorch, demonstrate its architecture and core functionalities, and apply it to a suitable industrial application to evaluate its effectiveness.
3. Using Scikit-learn, develop a model on a load _digits dataset, apply feature scaling, Predict on test set &evaluate performance
4. Explain briefly TensorFlow Architecture, & Pytorch Architecture.
5. Explain how Docker helps maintain consistency across different environments in deploying a fraud detection system.
6. Demonstrate hyperparameter **tuning** using GridSearchCV in Scikit-learn .
7. Apply popular hyperparameter tuning techniques (such as Grid Search, Random Search, and Bayesian Optimization) to optimize a machine learning model and analyze their effectiveness.
8. Apply model serialization techniques (using Pickle or Joblib) to save and reload a trained machine learning model for deployment.
9. Develop and apply a Flask RESTful API to serve a trained machine learning model for real-time predictions.
10. Demonstrate how Docker can simplify the deployment of a machine learning-based fraud detection system across different environments (e.g., development, staging, and production).
11. Explain how deploying a fraud detection model on AWS or Azure can help in automatically scaling the system based on transaction volume. Support your answer with a relevant use case.
12. Deploy a FastAPI application locally, integrate it with ngrok to expose the API to the public internet, and demonstrate real-time access and testing of the endpoints
13. Apply a step-by-step Docker workflow to containerize and deploy a Flask ML inference service: create a Dockerfile, build and tag the image, run the container with port mapping and env-vars, and validate the API.
14. Practical Benefits of Fraud Detection System model deployments with different deployment platforms (Stan alone Anaconda Framework, NGROK, Docker, Cloud).
15. Comparison of Fraud Detection System model deployments aspects with Docker & Cloud and Without Docker/Cloud.
16. Train and serialize a machine learning model using joblib or pickle, then load it back to make predictions on test data.
17. Deploy a trained ML model on a cloud platform (AWS or Azure), and verify predictions through an API call from a local machine.
18. Illustrate the advantages of cloud deployment of machine learning models across various aspects (scalability, cost, accessibility, and reliability) with suitable examples.

19. Build a comparative evaluation dashboard that visualizes accuracy, inference time, and memory usage for two deployed models (Logistic Regression vs XGBoost) using matplotlib, based on real test results.

20. Compare the hardware acceleration performance of CPU, GPU, and TPU for deep learning model training in TensorFlow, and implement a small experiment to measure speed differences.

21. Build a comparative evaluation dashboard that visualizes accuracy, inference time, and memory usage for two deployed models (Logistic Regression vs XGBoost) using matplotlib, based on real test results.

22. Train and serialize a machine learning model using joblib or pickle, then load it back to make predictions on test data.

AMLIS Unit-4 Questions:

1. Apply recall and F1-score to evaluate a model's effectiveness for predicting whether patients have a disease.
2. Apply the confusion matrix to classify whether a loan approval model is effective, given predicted vs. actual outcomes.
3. Apply ROC-AUC to compare two classifiers and decide which model performs better.
4. Apply evaluation metrics to explain why a high accuracy score may still indicate poor model performance in imbalanced datasets.
5. Apply fairness evaluation to identify bias in a recruitment model that favors one gender over another.
6. A model for hiring candidates shows gender bias. Propose a methodology to detect and mitigate this bias while keeping performance intact.
7. Apply fairness-aware evaluation techniques to a loan approval dataset and describe how Equal Opportunity and Demographic Parity can be measured.
8. You observe that a sentiment analysis model is biased toward one demographic group. Suggest an experimental setup to identify and reduce such bias.
9. For an autonomous driving ML system, explain how ethical considerations influence the choice of evaluation metrics beyond accuracy.
10. Consider a healthcare ML model where misclassification can lead to life-threatening risks. Apply the concept of cost-sensitive learning to redesign the evaluation strategy.
11. A face recognition system performs better on lighter-skinned individuals than darker-skinned individuals. Apply bias detection metrics to evaluate this disparity and propose fairness interventions.
12. Suppose you are tasked with deploying a recommendation system. Apply societal impact analysis to evaluate the risks of echo chambers and propose mitigation strategies.
13. Using a healthcare diagnosis dataset, apply fairness-aware evaluation methods to assess model bias across different demographic groups. Explain, how **Equal Opportunity** and

Demographic Parity can be quantitatively measured and critically evaluate the implications of these metrics for equitable healthcare decision-making.

14. In the context of a fraud detection system used by a financial institution, where false negatives may lead to severe monetary losses and false positives can harm customer trust, apply the principles of cost-sensitive learning to redesign the model evaluation strategy.

AMLIS Unit-5 Questions:

1. Apply the architecture of GPT to generate text-based summaries for a given paragraph.
2. Apply BERT embeddings to classify a set of customer reviews into positive and negative categories.
3. Apply a use-case of generative AI in healthcare (e.g., drug discovery or medical report generation) and explain its workflow.
4. Apply Graph Neural Networks (GNNs) to model relationships in a simple social network of friends and suggest possible connections.
5. Apply Edge AI concepts to deploy a face detection model on a mobile phone and explain how it works in real time.
6. Apply AI techniques for climate modeling to predict temperature patterns for the next 10 years.
7. Apply the concept of transfer learning using a foundation model like BERT to improve performance on a sentiment analysis dataset. Justify why transfer learning is beneficial here.
8. Suppose you are tasked with building a medical chatbot using GPT-like models. Apply architectural features of foundation models to design a solution ensuring contextual understanding.
9. Apply knowledge of fine-tuning methods to design an experiment that adapts a generative AI model for domain-specific text generation (e.g., legal documents).
10. Construct a graph neural network for a recommendation system. Explain how GNNs can effectively model relationships between users and items compared to traditional methods.
11. Apply GNNs to a biological network dataset. Describe how node classification can help in drug discovery or disease prediction.
12. You are deploying an ML model on an IoT-based smart agriculture device. Apply strategies of model compression and quantization to make the deployment feasible on edge devices.
13. Apply the architecture of GPT-based language models to generate contextually relevant customer support responses from an e-commerce sample chat dataset.
14. Apply BERT embedding's to categorize news headlines into predefined topics such as politics, sports, technology, and entertainment.